

Simon Vutov

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EXPERIENCE

Software Engineer Intern - Deephaven

May 2026 - Aug 2026

TypeScript, React, Java, Python, Playwright, CI/CD

Remote

- Cut cross-browser end-to-end CI runtime from **53 to 22 minutes** by sharding Playwright suites across six jobs and merging reports while preserving coverage.
- Built a client-requested Java/React system enabling administrators to target maintenance banners across all UI dashboards.
- Delivered UI-wide 2D splitter-intersection resizing for deeply nested panels across three- and four-way crossings; validated with **9 automated drag tests**.

Firmware Developer - UW Orbital - Satellite Design Team

Apr 2026 - Present

C, Python, ARM, CMake, GoogleTest, pytest

Waterloo, ON

- Built a lossless C/Python binary logging protocol for satellite firmware and ground-station tooling, reducing representative logs from **271 to 69 bytes (75%)** with byte-identical decoding.
- Reduced alarm dequeue from $O(n)$ shifting to $O(1)$ pop in the firmware scheduler.

Software Engineer Intern - BitGo

Sept 2025 - Dec 2025

Backend, Security, Prometheus, Grafana, Feature Flags

Palo Alto, CA (Remote)

- Shipped blind-signing checks behind a feature flag in warning-only mode, emitting Grafana signals for staged rollout.
- The checks exposed **1,000+** verification failures; collaborated with a co-intern to resolve hundreds of false positives caused by codebase bugs.

Software Engineer Intern - GenerativeModels.AI

Jan 2025 - Apr 2025

Python, FastAPI, React, Next.js, FAISS, Airflow

Toronto, ON

- Built FAISS-backed semantic search over **70k documents** with sub-800ms responses.
- Dockerized Airflow ingestion for **900k product listings**.

PROJECTS

Invariant - C++ Numerical Computing Library | C++17, pybind11, NumPy, CMake

2026

- Built a C++17 numerical library with NumPy-compatible pybind11 bindings, QR least-squares solving, CMake installation, and **13-job** cross-platform CI.

CPPNN - C++ Neural Networks | C++, Image Classification, CMake

2025

- Implemented C++ image-classification networks, achieving **91.16%** MNIST test accuracy after five training epochs.

EDUCATION

Bachelor of Applied Science, Computer Engineering - University of Waterloo

Expected Apr 2029

Cumulative average: 86%; top 10% in ECE 250 Algorithms and Data Structures

Waterloo, ON

TECHNICAL SKILLS

Languages: C, C++, Python, TypeScript/JavaScript, Java, Rust, C#

Frameworks and data: React, FastAPI, REST APIs, FAISS, Airflow, pybind11, NumPy

Systems and tools: CMake, Docker, ARM, Playwright, CI/CD, Prometheus, Grafana